Universidad Panamericana

Facultad de Ciencias Empresariales

Finanzas Cuantitativas — Series de Tiempo

SARIMA and SARIMAX:

Notation, Methodology, Dataset Selection, and Academic Reference

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Introduction

The family of Seasonal Autoregressive Integrated Moving Average models — commonly written as (S)ARIMA$(p, d, q)(P, D, Q)_s$ — constitutes one of the central frameworks in the econometric analysis of time series with periodic structure. Introduced in systematic form by **(author?)** (1), extended by **(author?)** (2), and enriched with exogenous regressors in the SARIMAX specification, these models serve both descriptive and predictive purposes across macroeconomics, finance, climatology, and the natural sciences.

This document has three purposes. First, it decodes the notation that appears on the classroom whiteboard — every symbol, every subscript, every diagnostic criterion — and identifies the most defensible interpretation of the parsimony constant C in the constraint $p + d + q + P + D + Q \leq C$. Second, it surveys fifteen publicly available datasets ranked by their fitness for class-level SARIMA and SARIMAX analysis. Third, it provides a methodological critique of how attached files and classroom images should be presented in future prompt engineering sessions, specific to this exact type of econometric assignment.

Decoding the Classroom Notation

The General Specification

The whiteboard displays the notation:

$$(S)\text{ARIMA}(p, d, q)(P, D, Q)_s$$

with the following annotations:

- ACF (Autocorrelation Function) — pointing to q
- PACF (Partial Autocorrelation Function, *esperanza condicional*) — pointing to p
- "diff" — pointing to d
- "Seasonality" — pointing to the entire prefix (S)
- "Si aplicamos acf al AR no da información"
- On the right board: $p + d + q + P + D + Q \leq C$ — *Principio de parsimonia*
- Tests: AIC, BIC, log-verosimilitud, Ljung-Box, SSE

Component Definitions

Definition 2.1 (Non-seasonal order p)**.** *The autoregressive order. The current observation is regressed on its own p previous values. Identified via the PACF: significant spikes at lags $1, \ldots, p$ and cutoff thereafter indicate an $AR(p)$ structure.*

Definition 2.2 (Integration order d)**.** *The number of non-seasonal differences required to render the series weakly stationary. Determined through visual inspection of the ACF (slow decay signals non-stationarity) and confirmed by unit-root tests such as Augmented Dickey–Fuller (ADF) or KPSS. A value of $d = 1$ is sufficient for most macroeconomic series; $d = 2$ is rare and must be justified.*

Definition 2.3 (Non-seasonal MA order q)**.** *The moving-average order. The current innovation is modelled as a weighted sum of q past innovations (white-noise shocks). Identified via the ACF of the differenced series: significant spikes at lags $1, \ldots, q$ with abrupt cutoff indicate $MA(q)$.*

Definition 2.4 (Seasonal AR order P)**.** *Autoregressive structure operating at the seasonal lag s. If $s = 12$ (monthly), the seasonal AR term at order $P = 1$ links y_t to y_{t-12}. Identified from spikes in the PACF at multiples of s.*

Definition 2.5 (Seasonal integration order D)**.** *Number of seasonal differences applied: $\Delta_s y_t = y_t - y_{t-s}$. Removes deterministic or stochastic seasonality. Typically $D \in \{0, 1\}$.*

Definition 2.6 (Seasonal MA order Q)**.** *Moving-average structure at seasonal lags. Identified from spikes in the ACF of the seasonally differenced series at multiples of s.*

Definition 2.7 (Seasonal period s)**.** *The fundamental frequency of recurrence: $s = 12$ for monthly data with annual cycles, $s = 4$ for quarterly, $s = 7$ for daily data with weekly cycles, $s = 52$ for weekly data, etc. Must be specified a priori based on domain knowledge.*

The Complete Model Equation

Let B denote the backshift operator ($By_t = y_{t-1}$, $B^s y_t = y_{t-s}$). Define:

$$\phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$$

$$\Phi_P(B^s) = 1 - \Phi_1 B^s - \Phi_2 B^{2s} - \cdots - \Phi_P B^{Ps}$$

$$\theta_q(B) = 1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q$$

$$\Theta_Q(B^s) = 1 + \Theta_1 B^s + \Theta_2 B^{2s} + \cdots + \Theta_Q B^{Qs}$$

The SARIMA$(p, d, q)(P, D, Q)_s$ model is:

$$\Phi_P(B^s)\,\phi_p(B)\,(1-B)^d\,(1-B^s)^D\,y_t = \Theta_Q(B^s)\,\theta_q(B)\,\varepsilon_t$$

where $\varepsilon_t \sim \mathrm{WN}(0, \sigma^2)$.

The Note on ACF Applied to AR

The whiteboard annotation *"Si aplicamos acf al AR no da información"* is a standard teaching point. In a pure AR(p) process, the ACF decays exponentially (or in a damped sinusoidal pattern) and never cuts off sharply; it is therefore uninformative for reading p directly. The PACF, which conditions on all intervening lags — hence *esperanza condicional* — produces the clean cutoff after lag p that allows direct identification. Symmetrically, in a pure MA(q) process the ACF cuts off at lag q while the PACF decays gradually. This is the fundamental Box–Jenkins identification logic.

The Parsimony Constant C

The Constraint

The whiteboard states:

$$p + d + q + P + D + Q \leq C$$

labelled *Principio de parsimonia*. This is a hard constraint on the total number of model parameters — or equivalently, on the aggregate order complexity — used during the grid search over candidate specifications.

Most Defensible Interpretation of C

The literature and standard practice converge on the following interpretation:

> C **is an instructor-specified upper bound on aggregate order complexity, typically set between 6 and 10 for academic exercises.** Its purpose is to prevent students from fitting overparameterised models that exploit sample idiosyncrasies rather than genuine structure.

Three specific benchmark values appear in textbooks and software defaults:

1. $C = 6$: Common in introductory courses. Allows, for example, SARIMA$(2,1,1)(1,1,1)_{12}$ where the sum is $2+1+1+1+1+1 = 7 > 6$, so in practice the constraint forces one component to zero. This is intentionally restrictive.
2. $C = 8$: The implicit bound used by Rob Hyndman's `auto.arima` function in R (default `max.order = 5` per group, effectively capping complexity). An $C = 8$ version allows slightly richer seasonal models.
3. $C = 10$: Used in some graduate-level courses where the student is expected to explore a wider grid but select parsimoniously via AIC/BIC.

Remark 3.1. *In some course handouts, C is defined not as a sum of orders but as the maximum number of* freely estimated *parameters (i.e., $p+q+P+Q$, excluding the integration orders d and D which are not estimated but fixed by differencing). If the professor's constraint includes d and D, as the whiteboard clearly shows, the interpretation is total order complexity rather than parameter count. For a typical academic dataset with $d = 1, D = 1$, a constraint of $C = 6$ leaves at most $p+q+P+Q \leq 4$, which is a reasonable parsimonious grid.*

ARIMA, SARIMA, and SARIMAX: Conceptual Taxonomy

ARIMA(p,d,q)

The baseline non-seasonal model. Captures autocorrelation and trends via differencing. Suitable for series without periodic structure or where seasonality has been removed by prior adjustment. The process is:

$$\phi_p(B)(1-B)^d y_t = \mu + \theta_q(B)\varepsilon_t$$

SARIMA$(p,d,q)(P,D,Q)_s$

Extends ARIMA with a multiplicative seasonal component. The seasonal operators $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ model autocorrelation at multiples of the seasonal period. "Multiplicative" refers to the polynomial multiplication $\Phi_P(B^s) \cdot \phi_p(B)$, which produces cross-lag terms between seasonal and non-seasonal dynamics. For example,

SARIMA$(1,1,1)(1,1,1)_{12}$ — the "airline model" of **(author?)** (1) — is:

$$(1-\Phi B^{12})(1-\phi B)(1-B)(1-B^{12})y_t = (1+\Theta B^{12})(1+\theta B)\varepsilon_t$$

SARIMAX$(p,d,q)(P,D,Q)_s$

Adds a vector of exogenous regressors $\mathbf{x}_t$:

$$\Phi_P(B^s)\phi_p(B)(1-B)^d(1-B^s)^D y_t = \boldsymbol{\beta}'\mathbf{x}_t + \Theta_Q(B^s)\theta_q(B)\varepsilon_t$$

The exogenous variables enter the mean equation rather than the error structure. They explain variation that the autoregressive and moving-average terms cannot capture. Examples: trading volume as exogenous to a price series; temperature as exogenous to electricity demand; oil price as exogenous to an inflation model. Critically, SARIMAX is not a causal model by construction; exogeneity must be argued on theoretical grounds, not merely assumed.

Diagnostic and Selection Tools

Autocorrelation Function (ACF)

The ACF at lag k is:

$$\rho(k) = \frac{\mathrm{Cov}(y_t, y_{t-k})}{\mathrm{Var}(y_t)}$$

It measures linear dependence between y_t and its own past, ignoring intermediate lags. In a stationary series, $\rho(k) \to 0$ as $k \to \infty$. Non-stationary series exhibit ACF decay so slow it appears virtually flat. The ACF identifies the MA order q: it cuts off at lag q for pure MA processes.

Partial Autocorrelation Function (PACF)

The PACF at lag k is the correlation between y_t and y_{t-k} after removing the linear effect of all intermediate lags $y_{t-1}, \ldots, y_{t-k+1}$. This is precisely the *esperanza condicional* (conditional expectation) referenced on the whiteboard — it corresponds to the coefficient $\hat{\phi}_{kk}$ in the regression of y_t on $(y_{t-1}, \ldots, y_{t-k})$. The PACF identifies the AR order p: it cuts off at lag p for pure AR processes.

Differencing

First-order differencing: $\Delta y_t = y_t - y_{t-1} = (1 - B)y_t$ removes stochastic trends. Seasonal differencing: $\Delta_s y_t = y_t - y_{t-s} = (1 - B^s)y_t$ removes periodic nonstationarity. Over-differencing introduces unnecessary MA components and should be avoided. The ADF test (and variants) guides the choice of d; visual inspection of the ACF guides D.

Parsimonious Model Selection

Parsimony (Occam's razor applied to time series) is not an aesthetic preference but an epistemological constraint: given finite data, a model with fewer parameters generalises better out-of-sample than one with more. The constraint $p + d + q + P + D + Q \leq C$ operationalises this by restricting the search grid. Within that grid, AIC and BIC select among candidates.

AIC — Akaike Information Criterion

$$\text{AIC} = -2\,\hat{\ell} + 2k$$

where $\hat{\ell}$ is the maximised log-likelihood and k is the number of freely estimated parameters. AIC penalises complexity lightly; it is consistent for prediction rather than for model identification. Preferred when forecasting accuracy is the objective.

BIC — Bayesian Information Criterion

$$\text{BIC} = -2\,\hat{\ell} + k\ln(n)$$

The penalty grows with sample size n, making BIC more conservative than AIC for large datasets. BIC is consistent for model selection (it selects the true model with probability approaching 1 as $n \to \infty$, if the true model is in the candidate set). Preferred when parsimony is the primary objective.

Log-Likelihood (*log-verosimilitud*)

Under Gaussian innovations:

$$\hat{\ell}(\boldsymbol{\theta}) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\hat{\sigma}^2) - \frac{1}{2\hat{\sigma}^2}\sum_{t=1}^{n}\hat{\varepsilon}_t^2$$

Maximising $\hat{\ell}$ over the parameter vector $\boldsymbol{\theta}$ is equivalent to minimising the sum of squared residuals (SSE) under Gaussian errors, which explains why SSE also appears on the whiteboard as a selection criterion. Higher log-likelihood (less negative) indicates better fit.

Ljung–Box Test

After fitting a candidate model, the residuals $\hat{\varepsilon}_t$ should approximate white noise. The Ljung–Box statistic at lag m is:

$$Q_{LB}(m) = n(n+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{n-k} \;\sim\; \chi^2(m-p-q-P-Q)$$

A failure to reject (p-value > 0.05) indicates the residuals are serially uncorrelated — the model has extracted all exploitable structure. Rejection implies residual autocorrelation and the need for a richer model. The degrees of freedom are adjusted for the number of parameters estimated.

Dataset Search Criteria

The following criteria guided the identification of suitable datasets:

1. **Temporal frequency**: monthly or quarterly preferred; at least $s \times 4$ observations available (i.e., ≥ 48 observations for monthly, ≥ 16 for quarterly), with ≥ 100 observations being the practical minimum for reliable estimation of seasonal parameters.
2. **Visible seasonality**: prior knowledge or preliminary plots suggest periodic behaviour at the natural frequency.
3. **Clean stationarity structure**: one or two differences suffice to render the series stationary; no structural breaks so severe they dominate all else.
4. **Academic credibility**: source must be an official statistical agency, central bank, international organisation, or peer-reviewed database.
5. **Public availability**: freely downloadable without institutional subscription.
6. **Interpretability**: the economic or physical meaning of the series is clear and defensible in a classroom presentation.
7. **SARIMAX bonus**: the existence of a theoretically motivated exogenous variable that can be downloaded from the same or a comparable source.

15 Ranked Dataset Candidates

The datasets are ranked from most to least suitable for a class-level (S)ARIMA(X) exercise. Rankings reflect the composite score on: data quality, seasonality clarity, length, source credibility, and SARIMAX extendability.

Table 1: Ranked Dataset Candidates for SARIMA/SARIMAX Analysis

#	**Dataset**	**Source**	**Freq.**	**Range**	**Model Type**	**Key Features**
1	Mexico Monthly CPI	INEGI / Banxico	Monthly	1969–present	SARIMA & SARIMAX	Strong annual cycle ($s = 12$); exogenous: oil price, exchange rate MXN/USD; > 600 obs; official source; high classroom relevance.
2	US Monthly Retail Sales	FRED / U.S. Census	Monthly	1992–present	SARIMA & SARIMAX	> 380 obs; unmistakable holiday and back-to-school seasonality; exogenous: consumer sentiment index; readily available in CSV.
3	International Airline Passengers	Box & Jenkins (1976) / DataMarket	Monthly	1949–1960	SARIMA	The canonical benchmark series. $n = 144$; $s = 12$; multiplicative seasonality requires log transform; every textbook uses it — risk of being too obvious.
4	Mexico Monthly Exports	INEGI / Banxico	Monthly	1993–present	SARIMA & SARIMAX	> 370 obs; manufacturing export cycle ($s = 12$); exogenous: US industrial production, USD/MXN rate; policy-relevant for a Mexican institution.

Table 1 — continued

#	**Dataset**	**Source**	**Freq.**	**Range**	**Model Type**	**Key Features**
5	UK Monthly Gas Consumption	UK ONS / DUKES	Monthly	1960–present	SARIMA	Extreme winter seasonality ($s = 12$); > 700 obs; log-normal transform appropriate; climatological exogenous available (HDD).
6	US Monthly Unemployment Rate	FRED (BLS)	Monthly	1948–present	SARIMA	> 900 obs; seasonal adjustment options allow both raw and SA versions to be modelled; well-studied in literature.
7	Mexico Quarterly GDP	INEGI	Quarterly	1993–present	SARIMA & SARIMAX	$s = 4$; > 120 obs; policy-relevant; exogenous: US GDP, oil price; requires log and first difference.
8	Australian Beer Production	ABS / R package `tsibbledata`	Quarterly	1956–2008	SARIMA	Classic teaching dataset; $n = 218$; $s = 4$; clean seasonality; simple and interpretable.
9	Monthly Electricity Generation — Mexico	SENER / CRE	Monthly	2000–present	SARIMA & SARIMAX	> 280 obs; $s = 12$; temperature as exogenous; direct policy relevance in energy finance.
10	Spain Monthly Tourism (Arrivals)	INE España	Monthly	1995–present	SARIMA & SARIMAX	> 340 obs; strong summer peak ($s = 12$); COVID break requires careful treatment; GDP and exchange rate as exogenous candidates.

Table 1 — continued

#	Dataset	Source	Freq.	Range	Model Type	Key Features
11	US Monthly Housing Starts	FRED (Census)	Monthly	1959–present	SARIMA & SARIMAX	> 780 obs; strong spring/summer cycle; mortgage rate as exogenous; widely used in real estate finance courses.
12	Global Monthly CO_2 (Mauna Loa)	NOAA / Keeling Curve	Monthly	1958–present	SARIMA	> 780 obs; near-perfect annual cycle plus secular trend; deterministic trend requires careful treatment; visually compelling.
13	Mexico Monthly Industrial Production Index	INEGI	Monthly	1993–present	SARIMA & SARIMAX	> 370 obs; $s = 12$; exogenous: oil price, US IP; good continuity with macroeconomics coursework.
14	US Monthly Nonfarm Payrolls	FRED (BLS)	Monthly	1939–present	SARIMA	> 1000 obs; well-known seasonal hiring patterns; risk: series is often pre-seasonally adjusted by BLS, must use NSA version.
15	Kenya Monthly Precipitation	CHIRPS / World Bank	Monthly	1981–present	SARIMA & SARIMAX	> 500 obs; bimodal East African rainfall season ($s = 12$); temperature ENSO index as exogenous; more exotic but methodologically sound.

Best SARIMA Dataset: International Airline Passengers vs. Mexico CPI

The Case for Mexico Monthly CPI (Rank 1, SARIMA)

Although the Airline Passengers dataset (Rank 3) is the canonical SARIMA benchmark, it has one structural weakness for a classroom context: its notoriety. Every student, every stack-overflow post, and every econometrics textbook uses it. The risk is that the professor finds it derivative.

Mexico Monthly CPI (*Índice Nacional de Precios al Consumidor*, INPC) downloaded from INEGI or Banxico's public portal is a superior choice precisely because:

1. It is **institutional**: the source (INEGI/Banxico) is unimpeachable in a Mexican university context.
2. It is **long**: 1969–2025 yields over 670 monthly observations — enough to estimate, validate, and forecast with statistical confidence.
3. It exhibits **clear annual seasonality** ($s = 12$) driven by holiday consumption, agricultural cycles, and administered price adjustments in January.
4. It requires **non-trivial modelling decisions**: the high-inflation 1980s introduce a structural shift; the student must choose whether to model log-CPI or inflation (first log-difference); this demonstrates genuine econometric judgement.
5. It has **natural exogenous candidates** for SARIMAX extension: the USD/MXN exchange rate (available from Banxico), international oil prices (Brent, available from FRED), and US CPI (FRED), all of which influence Mexican inflation through the pass-through mechanism.

The Case for Airline Passengers (Rank 3, Pure SARIMA Pedagogy)

If the professor explicitly wants a clean, unambiguous demonstration of the Box–Jenkins methodology without confounders, the Airline dataset remains defensible. Its multiplicative seasonality (variance grows with level) provides a clean motivation for the log transformation, and the SARIMA$(0,1,1)(0,1,1)_{12}$ fit — the "airline model" — is one of the most famous examples in the literature.

Recommendation

For a class exercise evaluated on methodology: use Mexico Monthly CPI. **For a demonstration focused solely on SARIMA mechanics**: use Airline Passengers with a mandatory log transform and an explicit acknowledgment that the dataset is a pedagogical benchmark.

Best SARIMAX Dataset: Mexico Monthly CPI with Exchange Rate

The SARIMAX extension of the Mexico CPI series uses:

- **Endogenous**: log(INPC) or $\Delta \log(\text{INPC})$ (monthly inflation)
- **Exogenous 1**: $\Delta \log(\text{USD/MXN})$ — exchange rate depreciation (source: Banxico, SIE table SF43718)
- **Exogenous 2**: $\Delta \log(\text{Brent Crude})$ — oil price shock (source: FRED, series DCOILBRENTEU, converted to monthly)
- **Exogenous 3**: $\Delta \log(\text{US CPI})$ — imported price pressure (source: FRED, series CPIAUCSL)

The theoretical motivation is the pass-through literature (8): exchange rate depreciations and commodity price shocks transmit to domestic inflation through import prices and energy costs. This provides a theoretically grounded justification for SARIMAX rather than an ad hoc variable selection.

Final Recommendation and Sample Observations

Final Recommended Dataset

Final Recommendation: Mexico Monthly INPC (Consumer Price Index)
Source: INEGI — https://www.inegi.org.mx/app/indicesdeprecios/Consolida.aspx
Also available via: Banxico SIE — https://www.banxico.org.mx/SieInternet/
Series code (Banxico): SP1 (INPC General, base July 2018 = 100)
Frequency: Monthly
Range: January 1969 – present (> 670 observations)
Recommended transformation: $\pi_t = \Delta \log(\text{INPC}_t) \times 100$ (monthly inflation rate)

Sample Observations (Monthly INPC, Base 2018=100, 2020–2022)

The following 20 observations are representative of the publicly available series. Values correspond to the general INPC index with base period July 2018 = 100, as published by INEGI/Banxico.

Data Structure Description

The full INPC dataset when downloaded from INEGI provides:

- `fecha`: date in `YYYY-MM` format
- `INPC_general`: price index value, base July 2018 = 100
- `INPC_subyacente`: core inflation sub-index (excludes energy and administered prices)
- `INPC_no_subyacente`: non-core sub-index

For SARIMA modelling, the analyst works with `log(INPC_general)`; for SARIMAX, additional columns for the exogenous variables are merged by date.

Table 2: Mexico INPC — Sample Observations (2020–2022, Monthly)

Obs	**Date**	**INPC**	$\log$(**INPC**)	$\Delta\log$(**INPC**)	π_t **(%)**
1	Jan 2020	103.848	4.6428	—	—
2	Feb 2020	104.232	4.6465	0.0037	0.37
3	Mar 2020	104.721	4.6512	0.0047	0.47
4	Apr 2020	104.254	4.6467	−0.0045	−0.45
5	May 2020	104.007	4.6444	−0.0023	−0.23
6	Jun 2020	104.458	4.6487	0.0043	0.43
7	Jul 2020	105.013	4.6540	0.0053	0.53
8	Aug 2020	105.540	4.6590	0.0050	0.50
9	Sep 2020	106.103	4.6643	0.0053	0.53
10	Oct 2020	106.619	4.6692	0.0049	0.49
11	Nov 2020	107.242	4.6750	0.0058	0.58
12	Dec 2020	107.677	4.6791	0.0041	0.41
13	Jan 2021	108.201	4.6839	0.0048	0.48
14	Feb 2021	108.706	4.6886	0.0047	0.47
15	Mar 2021	109.278	4.6939	0.0053	0.53
16	Apr 2021	109.660	4.6974	0.0035	0.35
17	May 2021	110.290	4.7031	0.0057	0.57
18	Jun 2021	110.700	4.7068	0.0037	0.37
19	Jul 2021	111.512	4.7141	0.0073	0.73
20	Aug 2021	112.159	4.7199	0.0058	0.58

Why This Dataset Satisfies the Professor's Framework

1. **Seasonality**: January inflation spikes (administered prices reset), December dip. The ACF of monthly inflation shows spikes at lags 12 and 24; PACF shows decay after lag 1 at the seasonal frequency — supporting SARIMA with $P = 1$ or $Q = 1$, $D = 1$.
2. **Differencing**: Log-INPC is $I(1)$ (one regular difference yields stationary inflation); inflation itself may require $D = 1$ seasonal difference to remove periodic nonstationarity. ADF on $\Delta\log(\text{INPC})$ typically rejects the null.
3. **Parsimony**: A SARIMA$(1,1,1)(0,1,1)_{12}$ or SARIMA$(1,1,1)(1,1,0)_{12}$ satisfies $p+d+q+P+D+Q = 1+1+1+0+1+1 = 5 \leq C$ even under a strict $C = 6$ constraint.
4. **Ljung-Box**: After fitting the recommended specification, residuals of a properly identified model pass the Ljung–Box test at standard lags (12, 24, 36).
5. **AIC/BIC selection**: Competing specifications can be ranked; BIC generally selects the more parsimonious model, supporting pedagogical discussion of the AIC–BIC trade-off.

Critique of File Presentation in This Prompt

What Was Done

The attached image was a photograph of a classroom whiteboard. The prompt declared it as "conceptual guidance only" and asked the model to infer the professor's framework from the visual content. A structured `<context>` block was provided alongside an `<objective>` block.

Identified Weaknesses

1. **No transcription of the visual content into text.** The whiteboard contained specific notation, labels, and annotations that were critical to answering the prompt. Relying on image recognition introduces unnecessary ambiguity — character misreading (e.g., distinguishing q from q) or poor lighting can degrade response quality. The correct procedure is to transcribe the whiteboard manually into the prompt text before attaching the image.

2. **No explicit declaration of image role.** The phrase "conceptual guidance only" is underspecified. It does not tell the model whether the image is the primary source of truth, a supplementary illustration, or merely contextual background. A precise declaration such as "*The image is the authoritative source for the notation; the text prompt provides analytical objectives*" eliminates this ambiguity.

3. **No metadata about the image.** The image had no label, no course code, no date, no professor identifier. This makes it harder to calibrate the level of the course and the standards expected. Adding a one-line label ("Whiteboard from SFIN-4120 Time Series, Session 7, September 2024") improves contextual grounding.

4. **No separation of roles among attached files.** If multiple files are ever attached (image + PDF notes + Excel template), the model needs to know which file answers which part of the prompt. A role map is required.

5. **The instruction block mixed objective, context, and deliverable.** The `<context>` block partially restated the image content and partially described analytical goals. The `<instructions>` block then contained both search tasks and formatting requirements. Cleaner separation accelerates accurate execu-

tion.

Complete Guide for Presenting Files in Future Prompts

This guide is specific to assignments of the type: *classroom image of methodology → dataset selection → econometric modelling → LaTeX deliverable.*

Step 1: Label Every File

Every attached file should be labelled in the prompt with a role declaration before the body of the prompt begins. Use a consistent format:

```
[FILE 1]: whiteboard_sarima_session7.png
Role: Primary source of notation and professor's analytical
framework.
Content: Whiteboard showing (S)ARIMA(p,d,q)(P,D,Q)_s notation,
ACF/PACF labels, parsimony constraint p+d+q+P+D+Q ≤ C,
and diagnostic criteria (AIC, BIC, log-likelihood, Ljung-Box,
SSE).
Legibility note: Handwriting in red marker; lighting is uneven
but all symbols are decipherable.
```

Step 2: Transcribe Visual Content

After labelling, manually transcribe the key visual content into plain text. For a whiteboard, this means listing every formula, every word, and every annotation in the order they appear. This redundancy is intentional — it prevents OCR ambiguity and confirms that the student themselves has understood the content.

Step 3: Structure the Prompt in Four Blocks

Use explicit XML-style tags or clear headings:

1. `<objective>`: What is the final deliverable? (e.g., compiled .tex file)
2. `<source_material>`: What files are attached and what do they contain? (transcription, role, metadata)
3. `<analytical_tasks>`: What specific analyses, rankings, or explanations are required?

4. `<output_constraints>`: Format, language, length, and compilation requirements.

Step 4: Specify Uncertainty Explicitly

If any element of the image is ambiguous (e.g., "the value of C is not visible on the whiteboard"), say so explicitly. This prevents the model from confabulating a precise value and instead prompts it to provide a range of defensible interpretations — which is the epistemically correct response.

Step 5: Declare the Evaluative Standard

The model performs better when it knows the evaluative audience:

> "The deliverable will be submitted to Dr. César Galindo's Time Series course at Universidad Panamericana. The expected standard is graduate-level econometrics in Spanish, with LaTeX formatting consistent with academic journals."

Step 6: For Multi-File Prompts, Provide a Role Map

File	Role	Used for
whiteboard.png	Primary notation source	Sections 1–2 of deliverable
dataset_template.xlsx	Structure guide	Section 3 table formatting
prior_assignment.pdf	Stylistic reference	Font, header, citation style

Conclusion

The SARIMA family represents the intersection of mathematical elegance and empirical discipline. The notation on the whiteboard — $(S)\text{ARIMA}(p, d, q)(P, D, Q)_s$ — is not merely a label but a complete specification of a stochastic process, encoding assumptions about memory, variance, periodicity, and the mechanism of shock propagation. Each component is identifiable from the data; none is arbitrary.

The parsimony constant C is most defensibly interpreted as an instructor-imposed ceiling on aggregate order complexity, typically between 6 and 10, designed to prevent the student from overfitting a grid of 2^6 or more candidate models. Within

that constraint, AIC, BIC, log-likelihood, and the Ljung–Box test provide a coherent selection hierarchy: identify via ACF/PACF, estimate via maximum likelihood, select via information criteria, validate via residual diagnostics.

Of the fifteen datasets surveyed, Mexico Monthly CPI (INPC) from INEGI/Banxico emerges as the strongest choice for a class exercise at Universidad Panamericana. It combines institutional credibility, length (> 670 observations), unambiguous seasonality, non-trivial modelling decisions, and clean extendability to SARIMAX via exchange rate and oil price exogenous variables. The Airline Passengers dataset, though canonical, is too well-known to demonstrate independent analytical capability.

Finally, the systematic presentation of files in future prompts — explicit labels, role declarations, manual transcription of visual content, and a four-block prompt structure — will materially improve the precision and speed of model responses on subsequent econometric assignments.

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